Chapter 15, Problem 1PP

Problem

Find the Laplace transforms of these functions: *r*(*t*) = *tu*(*t*), that is, the ramp function; *Ae−at u*(*t*); and *Be−jωtu*(*t*)

Step-by-step solution

Step 1 of 3

The ramp signal $r(t) = tu(t)$ and is defined as,

$$\begin{aligned} r(t) &= t \quad \text{when } u(t) = 1 \text{ for } t \geq 0, \\ &= 0 \quad \text{when } u(t) = 0 \text{ for } t < 0. \end{aligned}$$

The Laplace transform of the ramp signal is,

$$\begin{aligned} \mathcal{L}(tu(t)) &= \int_0^\infty te^{-st}\,dt \\ &= t\frac{e^{-st}}{-s}\bigg|_0^\infty - \int_0^\infty \frac{e^{-st}}{-s}\,dt \\ &= \frac{-1}{s}\left(\infty \cdot e^{-\infty} - 0 \cdot e^{-0}\right) + \frac{e^{-st}}{-s^2}\bigg|_0^\infty \\ &= \frac{-1}{s}(0-0) - \frac{1}{s^2}\left(e^{-\infty} - e^{-0}\right) \end{aligned}$$

$$= 0 - \frac{1}{s^2}(0-1)$$

$$= \frac{1}{s^2}$$

Therefore, the Laplace transform of the ramp signal $tu(t)$ is $\boxed{\frac{1}{s^2}}$.

Step 2 of 3

The exponential signal is denoted as $s(t) = Ae^{-at}u(t)$ and is defined as,

$$s(t) = Ae^{-at} \quad \text{when } u(t) = 1 \text{ for } t \geq 0,$$
$$= 0 \quad \text{when } u(t) = 0 \text{ for } t < 0.$$

The Laplace transform of the exponential signal is,

$$\mathcal{L}\left(Ae^{-a}u(t)\right) = \int_0^{\infty} Ae^{-at}e^{-st}\,dt$$
$$= \int_0^{\infty} Ae^{-(s+a)t}\,dt$$
$$= A\left.\frac{e^{-(s+a)t}}{-(s+a)}\right|_0^{\infty}$$
$$= -\frac{A}{s+a}\left(e^{-(s+a)\infty} - e^{-(s+a)0}\right)$$
$$= -\frac{A}{s+a}(0-1)$$
$$= \frac{A}{s+a}$$

Therefore, the Laplace transform of $Ae^{-a}u(t)$ is $\boxed{\frac{A}{s+a}}$.

Step 3 of 3

The exponential signal is denoted as $q(t) = Be^{-j\omega t}u(t)$ and is defined as,

$$q(t) = Be^{-j\omega t} \quad \text{when } u(t) = 1 \text{ for } t \geq 0,$$
$$= 0 \quad \text{when } u(t) = 0 \text{ for } t < 0.$$

The Laplace transform of the exponential signal is,

$$\mathcal{L}\left(Be^{-j\omega t}u(t)\right) = \int_0^{\infty} Be^{-j\omega t}e^{-st}\,dt$$
$$= \int_0^{\infty} Be^{-(j\omega+s)t}\,dt$$
$$= B\left.\frac{e^{-(s+j\omega)t}}{-(s+j\omega)}\right|_0^{\infty}$$
$$= -\frac{B}{s+j\omega}\left(e^{-(s+j\omega)\infty} - e^{-(s+j\omega)0}\right)$$
$$= -\frac{B}{s+j\omega}(0-1)$$
$$= \frac{B}{s+j\omega}$$

Therefore, the Laplace transform of $Be^{-j\omega t}u(t)$ is $\boxed{\frac{B}{s+j\omega}}$.